

Lecture 4 — Power Factor

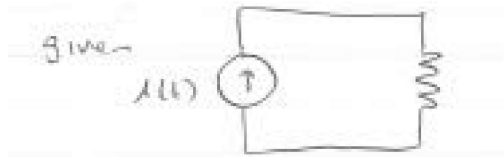
Scribe: MIT Student

1 Definitions and Identities

1. The RMS (root mean square) of a waveform):

$$X_{RMS} \triangleq \sqrt{\frac{1}{T} \int_{\langle T \rangle} X^2(t) dt}$$

This is useful:



$$p(t) = I(t)^2 R$$

$$\langle P \rangle = \frac{1}{T} \int I(t)^2 R dt = I_{RMS}^2 R$$

The RMS is the DC (current or voltage) that gives the same average power dissipation in a resistor as the time-varying waveform.

2. Two waveforms are orthogonal on $[a, b]$ iff

$$\int_a^b x(t)y(t) dt = 0$$

Now, as sinusoidal with different frequencies are orthogonal ...

$$\int_0^{2\pi} \sin(m\omega t) \sin(n\omega t + \theta) d(\omega t) = 0 \text{ if } n \neq m$$

And as sin and cos are orthogonal ...

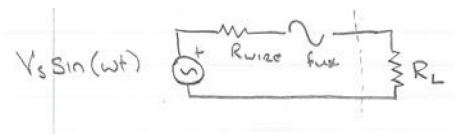
$$\int_0^{2\pi} \sin(\omega t) \cos(\omega t) = 0$$

In general ...

$$\frac{1}{2\pi} \int_0^{2\pi} \sin(\omega t) \sin(\omega t + \phi) d\omega t = \frac{1}{2} \cos \phi$$

2 Background

Suppose we plug a resistor into the wall

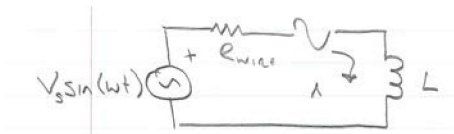


At 115 V_{RMS} , 15 A fuse, we can draw $p \simeq 1.7kW$ from wall (negatively R_{wire})

Neglectance R_{wire}

$$\begin{aligned} p &= \langle V(t)I(t) \rangle \\ &= \langle I(t)^2 R_L \rangle \\ &= I_{RMS}^2 R_L \end{aligned}$$

Suppose instead we plug an **inductor** into the wall



Neglectance R_{wire}

$$I = \int \frac{V}{L} dt = \frac{V_s}{\omega L} \cos \omega t$$

So

$$\langle P \rangle = -\frac{1}{2\pi} \frac{V_s^2}{\omega L} \int_0^{2\pi} \sin(\omega t) \cos(\omega t) d(\omega t) = 0$$

Because sine and cosine are orthogonal.

This makes sense, inductors store energy not dissipate it. (We slosh energy back and forth with the source.)

However, while no **average** power is drawn, there is instantaneous power (slosh energy into and out of wall) and RMS current drawn.

$$I_{RMS} < \frac{V_s}{\sqrt{2}\omega L}$$

At 115 V_{RMS} and 60Hz

$$L < 20mH$$

$$I_{RMS} > 15A$$

While we don't dissipate energy in the load, we **still** dissipate energy in R_{wire} ($I_{RMS} > 0$), and can blow the fuse.

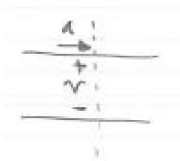
In this case, we are still not using the power capability of the source well.

To provide a measure of the utilization of the source, we define **Power Factor**:

$$P.F. \triangleq \frac{\langle P \rangle}{V_{RMS} I_{RMS}}$$

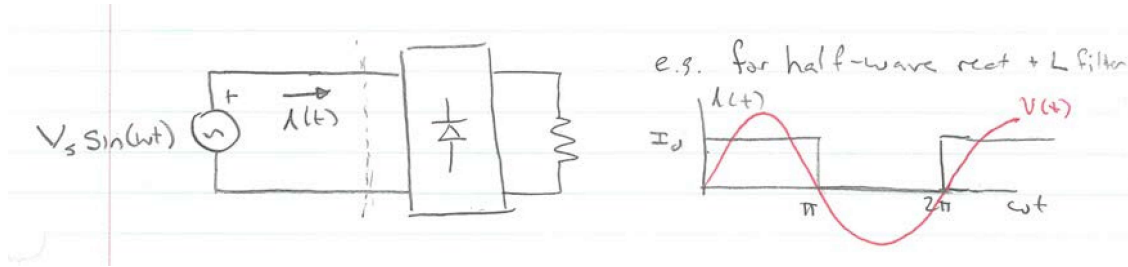
Where begin $\langle P \rangle$ represents the real power and $V_{RMS} I_{RMS}$ represents the apparent power.

For a resistor, $\langle P \rangle = V_{RMS} I_{RMS} \rightarrow P.F. = 1$ and an inductor $\langle P \rangle = 0 \rightarrow P.F. = 0$. Note that **we define power factor at a port**.



3 Rectifiers

Consider a rectifier drawing some current waveform.



In P.S.S., we can express $I(t)$ as a Fourier Series.

$$I(t) = \sum_{n=0}^{\infty} I_n \sin(n\omega t + \phi_n)$$

where $\phi_0 = 90^\circ$

For this, it is easy to show:

$$I_{RMS} = \sqrt{I_0^2 + \frac{I_1^2}{2} + \frac{I_2^2}{2} + \dots + \frac{I_n^2}{2} + \dots}$$

We can calculate power as:

$$\begin{aligned} \langle P \rangle &= \frac{1}{2\pi} \int_0^{2\pi} V_s \sin \omega t \sum_{n=0}^{\infty} I_n \sin(n\omega t + \phi_n) d(\omega t) \\ &= \sum_{n=0}^{\infty} V_s I_n \cdot \frac{1}{2\pi} \int_0^{2\pi} \sin(\omega t) \sin(\omega t + \phi_n) d(\omega t) \end{aligned}$$

From our orthogonality theorem, all but $n = 1$ term $\rightarrow 0$.

$$\langle P \rangle = V_s I_1 \cdot \frac{1}{2\pi} \int_0^{2\pi} \sin(\omega t) \sin(\omega t + \phi_1) d(\omega t)$$

From identity ...

$$\langle P \rangle = \frac{1}{2} V_s I_1 \cos \phi_1$$

or

$$\langle P \rangle = V_{RMS} I_{1,RMS} \cos \phi_1$$

So the only current that contributes to real power (average power) is the portion of the **fundamental current in phase with the voltage**!

Now:

$$P.F. = \frac{\langle P \rangle}{V_{RMS} I_{RMS}}$$

$$\begin{aligned}
&= \frac{V_{RMS} I_{1,RMS} \cos \phi_1}{V_{RMS} I_{RMS}} \\
&= \frac{I_{1,RMS} \cos \phi_1}{I_{RMS}}
\end{aligned}$$

Therefore,

$$P.F. = \frac{I_{1,RMS}}{I_{RMS}} \cos \phi_1 = k_d k_\theta$$

where $k_d = \frac{I_{1,RMS}}{I_{RMS}}$ is the distortion factor and $k_\theta = \cos \phi_1$ is the displacement factor.

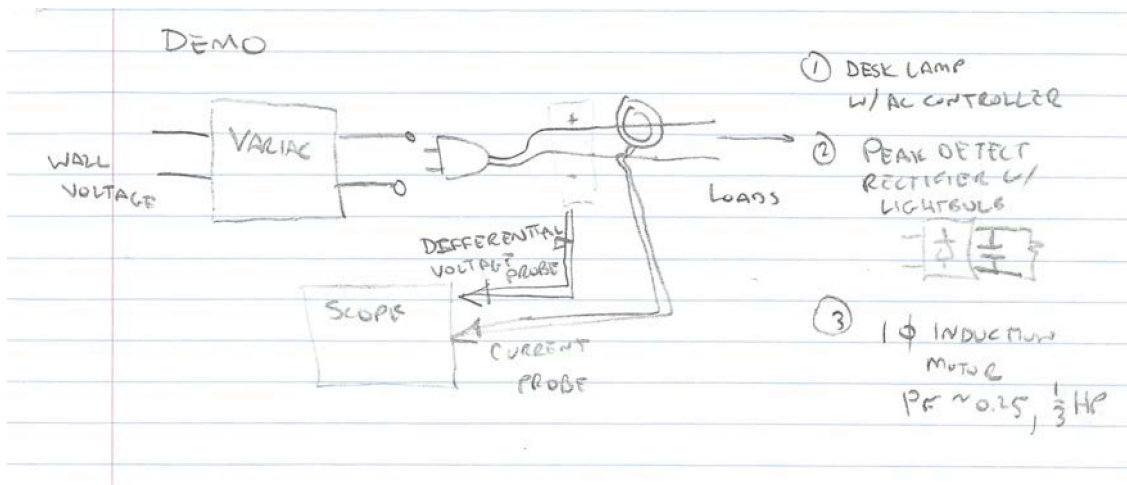
- k_d **Distortion factor** ($0 \leq k_d \leq 1$) tells us how much utilization is reduced because of harmonic currents that don't contribute real power.
- k_θ **Displacement factor** ($|k_\theta| \leq 1$) tells us how much utilization of the source is reduced because of phase shift between voltage and fundamental current.

Note:

1. Point out that since voltage is sinusoidal, due to orthogonality we only factor x for power from the fundamental current, and only from the component in phase with the voltage.
2. Breakdown and interpretation become more complex if both V and I are nonsinusoidal. Less usual case but we can still define P.F.

If time, introduce TDH (Total Harmonic Distortion) $THD \triangleq \sqrt{\frac{\sum_{n \neq 1} I_n^2}{I_1^2}}$

RMS of harmonic content normalized to RMS of fundamental. This assumes no DC content.



MIT OpenCourseWare
<https://ocw.mit.edu>

6.622 Power Electronics
Spring 2023

For information about citing these materials or our Terms of Use, visit: <https://ocw.mit.edu/terms>